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TRA-SAE: Low-Cost Retrieval-Augmented Fine-Tuning for Mixed-Domain Scientific Reasoning with a 4B Language Model — A Controlled Attribution Study

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ABSTRACT

Scientific and educational question answering couples numerical physical reasoning with symbolic logical inference, yet resource-constrained settings often preclude large proprietary models or extensive reinforcement learning. This paper reports TRA-SAE, a low-cost pipeline for the EXACT 2026 mixed-domain (physics and logic) reasoning challenge, built on a 4B-parameter open model. TRA-SAE couples retrieval-augmented few-shot inference with a three-stage curriculum: general supervised fine-tuning (SFT), a logic-focused curriculum stage, and Group Relative Policy Optimization (GRPO) with lightweight format, correctness, and unit-consistency rewards. Total training cost is US$18, and the full pipeline including all evaluations is below US$64 on a single A100-80GB. Every system is evaluated under a single greedy decoding pass. On a 217-sample validation split, the strongest configuration raises overall accuracy from 29.95% to 47.47% (63.12% physics, 18.42% logic). A controlled decomposition attributes +9.22 percentage points (pp) of the gain to retrieval-augmented exemplars and a further, borderline +5.07 pp to supervised fine-tuning, whereas the GRPO increment over the curriculum stage is not statistically significant. Under this single-pass protocol, retrieval—not weight training—supplies the majority of attainable gains at this scale, while GRPO reward shaping and a bespoke unit-consistency reward add no statistically separable benefit. A fair multi-prompt re-evaluation places five 7B-class baselines at 11.5–29.0%, in-domain fine-tuning transfers to external MMLU physics (+6.33 pp) and FOLIO logic (+13.80 pp), and a tool-augmented variant yields only a marginal change. A structured error analysis attributes most residual errors to answer-extraction failures and identifies formal logic as the principal reasoning bottleneck at this scale.

1 | Introduction

Large language models (LLMs) achieve strong results on mathematical and logical reasoning benchmarks (Wei et al., 2022; X. Wang et al., 2023; Kojima, Gu, Reid, Matsuo, & Iwasawa, 2022). Deploying capable models under resource constraints, where larger proprietary systems are unavailable, requires the joint design of the training-data curriculum, the reward specification, and the parameter-efficient adaptation strategy.

Scientific reasoning presents a combined challenge. Physics problems require formula selection, symbolic manipulation, and unit-consistent numerical computation, whereas logic problems require quantifier handling, propositional inference, and the detection of common fallacies. The two modalities favour different inductive biases, yet a single model is expected to address both.

EXACT 2026 is the 2nd International XAI Challenge for Transparent Educational Question-Answering, organised as an IEEE IJCNN 2026 competition (URA Research Group, 2026). It succeeds the XAI Challenge 2025 on explainable educational question answering over academic regulations (Nguyen et al., 2025) and broadens the scope from academic-policy logic to

STEM physics, so that a system must address both logic-based educational queries and physics problems on electric circuits and electrostatics. The task provides a controlled testbed with 1,945 training examples (1,213 physics, 732 logic) and a held-out evaluation set, with a reported public baseline of 38.71%. This study treats the task as a proxy for mixed-domain scientific and educational reasoning and quantifies how far a Qwen3.5-4B model can be advanced using publicly available fine-tuning techniques under a fixed compute budget. To support an unambiguous interpretation, every system in this study, fine-tuned and baseline, is evaluated under a single greedy decoding pass without any answer-key-dependent regeneration.
The contributions of this paper are the following.

- A low-cost three-stage curriculum pipeline (general SFT → logic curriculum SFT → GRPO reward shaping), combined with retrieval-augmented few-shot inference, for mixed-domain scientific question answering with a 4B open model, at a full end-to-end cost below US$64 on a single A100-80GB.
- An attribution of accuracy gains across pipeline stages under a uniform single-pass protocol: a controlled decomposition attributes +9.22 pp to retrieval-augmented exemplars ($p < 10^{-3}$) and a further, borderline +5.07 pp to supervised fine-tuning ($p = 0.054$), while the logic-curriculum and GRPO increments (+1.84 pp and +1.38 pp) are not individually significant at the validation-set size; these uncertainties are reported explicitly.
- A fair multi-prompt baseline re-evaluation in which each 7B-class model is evaluated under its native answer format, which separates prompt-format compliance from reasoning capability.
- An external-benchmark evaluation on MMLU physics and FOLIO logic that quantifies the transfer of in-domain fine-tuning, a retrieval-controlled re-evaluation of the open baselines, and a tool-augmented evaluation that quantifies the effect of Python and Z3 assistance.
- A characterisation of negative results (self-consistency voting and Dual-LoRA expert routing) and a structured error analysis that separates answer-extraction failures from substantive reasoning errors.

The remainder of the paper is organised as follows. Section 2 reviews related work. Section 3 formalises the task. Section 4 describes the TRA-SAE pipeline. Section 5 presents the experimental setup. Section 6 reports the results. Section 7 discusses limitations, and Section 8 concludes.

2 | Related Work

2.1 | Chain-of-thought and self-consistency

Wei et al. (2022) showed that step-by-step prompting improves LLM performance on arithmetic and symbolic reasoning, and Kojima et al. (2022) demonstrated zero-shot chain-of-thought prompting. X. Wang et al. (2023) proposed self-consistency decoding, which samples multiple reasoning paths and aggregates the most frequent answer. The present results indicate that self-consistency provides no net benefit at 4B scale on this task (Section 6).

2.2 | Mathematics and science language models

DeepSeekMath (Shao et al., 2024) combined large-scale mathematical pre-training with GRPO. Qwen2.5-Math (Yang et al., 2024) and the broader Qwen family (Qwen Team, 2025) extended this approach across model scales. Llemma (Azerbayev et al., 2024) performed continued pre-training on scientific and proof corpora. MetaMath (Yu et al., 2023) and MAmmoTH (Yue et al., 2023) constructed augmented mathematical reasoning datasets. The present work operates at a smaller scale (4B) in a two-domain setting (physics and logic) and introduces a unit-consistency reward term.

2.3 | Neuro-symbolic integration

Logic-LM (Pan, Albalak, Wang, & Wang, 2023) and LINC (Olausson et al., 2023) couple LLMs with formal solvers for logical reasoning, and PAL (Gao et al., 2022) delegates arithmetic to a Python interpreter. Math-Shepherd (P. Wang et al., 2024) and process-supervision methods (Lightman et al., 2023) provide step-level reward labels. This study integrates the Z3 SMT solver into the correctness reward and into a tool-augmented inference variant, while relying on outcome-level rewards to avoid step-level annotation.

2.4 | Parameter-efficient fine-tuning and reinforcement learning

LoRA (Hu et al., 2022) decomposes weight updates into low-rank matrices, and QLoRA (Dettmers, Pagnoni, Holtzman, & Zettlemoyer, 2023) extends it to quantised backbones; LoRAHub (Huang et al., 2023) composes multiple domain adapters. InstructGPT (Ouyang et al., 2022) established RLHF via PPO (Schulman, Wolski, Dhariwal, Radford, & Klimov, 2017); Direct Preference Optimization (Rafailov et al., 2023) removes the explicit reward model; and GRPO (Shao et al., 2024) computes relative rewards within a generation group, which makes single-GPU training of 4B-class models practical. Curriculum learning (Bengio, Louradour, Collobert, & Weston, 2009) motivates the staged data ordering used here.

2.5 | Explainable educational QA challenges

The closest predecessor to EXACT 2026 is the XAI Challenge 2025, an explainable educational question-answering competition organised by the URA Research Group at Ho Chi Minh City University of Technology together with the TRNS-AI workshop at IJCNN 2025 (Nguyen et al., 2025). That challenge required systems to produce not only final answers but also logic-grounded natural-language explanations for questions about university policies; its dataset (481 training and 50 test records) was constructed from first-order-logic templates with Z3 validation and expert-student review, with premises supplied in both natural language and first-order logic. EXACT 2026 continues this line of work and broadens it from academic-regulation logic to STEM physics, evaluating submissions along answer correctness, explanation quality, and reasoning depth while encouraging—but not requiring—symbolic tools such as Z3 (URA Research Group, 2026). In contrast to the 2025 challenge report, which presents the competition, dataset, and evaluation protocol, the present paper is a system-level empirical study: it fixes a single 4B open model and reports controlled ablations of retrieval, supervised fine-tuning, logic-curriculum adaptation, GRPO reward shaping, tool augmentation, and self-consistency. We do not propose a new training algorithm; the contribution is a controlled attribution of which standard components actually pay off at 4B scale and which do not, isolated by the retrieval-only control (cfg0-R) and a uniform single-pass protocol. Table 1 situates TRA-SAE relative to representative reasoning systems and to these two challenge artefacts.

3 | Problem Formulation

Let $\mathcal{D} = \mathcal{D}_{\text{phys}} \cup \mathcal{D}_{\text{logic}}$ denote the EXACT 2026 dataset (URA Research Group, 2026). Each sample $(q_i, a_i^*, s_i) \in \mathcal{D}$ comprises a question q_i, a ground-truth answer a_i^*, and a domain label $s_i \in \{\text{physics}, \text{logic}\}$. A model M with parameters θ maps a question to a free-text response $\hat{y}_i = M_\theta(q_i)$. A structured extraction operator recovers the candidate answer $a_i = \texttt{extract}(\hat{y}_i)$, and accuracy is

$$\text{Acc} = \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{i=1}^{|\mathcal{D}_{\text{val}}|} \mathbf{1}\big[\text{verify}(a_i, a_i^*, s_i, q_i)\big], \tag{1}$$

where verify$(\cdot)$ applies Z3-based symbolic checking, when a parseable formal representation is available, and SI-unit-normalised numerical comparison for physics samples. Each response is produced by a single greedy decoding pass; no component of the evaluation uses the ground-truth label to decide whether to regenerate a response. The official challenge scores submissions along three dimensions—answer correctness, explanation quality, and reasoning depth—and encourages optional structured evidence such as first-order-logic derivations and chain-of-thought steps (URA Research Group, 2026); the present study targets the answer-correctness dimension and additionally examines answer formatting and reasoning errors in Section 6.

4 | Methodology

Figure 1 summarises the pipeline. All stages apply LoRA (r=32, α=64) to Qwen3.5-4B.

4.1 | Structured response format

All models are instructed to produce a three-tag response, $\langle$`reasoning`$\rangle$ … $\langle$`/reasoning`$\rangle$$\langle$`answer`$\rangle$ … $\langle$`/answer`$\rangle$$\langle$`explanation`$\rangle$ … $\langle$`/explanation`$\rangle$, which supports reliable answer extraction and enables a format-based reward term.

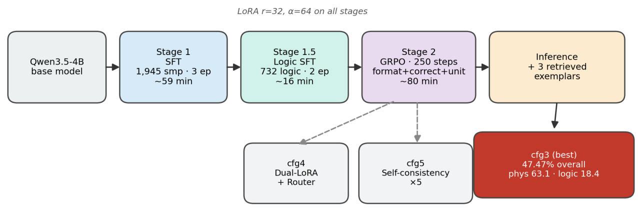

FIGURE 1 | Overview of the TRA-SAE training and inference pipeline. Stage 1 performs general supervised fine-tuning; Stage 2 applies a logic-focused curriculum stage; Stage 3 applies GRPO with a composite reward. At inference, the fine-tuned configurations retrieve three in-context exemplars. Optional Dual-LoRA routing and self-consistency voting are evaluated as additional configurations.

4.2 | Phase 1: supervised fine-tuning

Qwen3.5-4B is fine-tuned on the full training set (1,945 samples) with a cross-entropy loss on the target response. The configuration is three epochs, per-device batch size four with gradient accumulation four (effective batch 16), learning rate 2×10^{-4}, LoRA r=32, α=64, dropout 0.05, applied to all seven projection matrices. Training completes in 58.8 min and converges to a training loss of 0.3085.

4.3 | Phase 1.5: logic curriculum fine-tuning

A second SFT stage is restricted to the logic subset (732 samples) with a reduced learning rate of 5×10^{-5} for two epochs. This curriculum stage (Bengio et al., 2009) allocates additional capacity to formal reasoning, which is under-represented (37.6% of the training data). Training completes in 15.7 min and converges to a loss of 0.1326.

4.4 | Phase 2: GRPO with reward shaping

For a completion y with ground truth a^* and domain s, the reward is

$$\begin{aligned} R(y, a^*, s) = {} & w_f\, r_{\text{format}}(y) + w_c\, r_{\text{correct}}(y, a^*, s) \\ & + w_u\, r_{\text{unit}}(y, a^*) + \lambda\, \mathbf{1}[\, |y_{\text{reason}}| > 800\,], \end{aligned} \tag{2}$$

with w_f=0.30, w_c=0.60, w_u=0.10, and λ= − 0.10, where $|y_{\text{reason}}|$ is the reasoning token count. The format term awards 0.10 per present tag (maximum 0.30). The correctness term is binary and applies a $\leq$ 5% relative tolerance after SI-unit normalisation for physics and normalised string or Z3-verified matching for logic. The unit-consistency term equals one when the extracted answer carries a unit dimensionally compatible with the ground-truth unit. GRPO (Shao et al., 2024) is applied from the Phase-1.5 checkpoint with K=4 completions per prompt; the group-normalised advantage is $\hat{r}_k = (R_k - \bar{R})/(\sigma_R + \epsilon)$, with KL coefficient β=0.04, learning rate 10^{-6}, and 250 steps. Training completes in 79.8 min and reaches a loss of 0.0295.

TABLE 1 | Positioning of TRA-SAE relative to representative LLM reasoning systems and to the EXACT challenge artefacts. "Tuning" indicates whether the approach trains model weights.

Work	Focus / domain	Tuning	Relation to TRA-SAE
XAI Challenge 2025 report (Nguyen et al., 2025)	Academic-regulation QA	—	Predecessor; defines the challenge and dataset, not a system study
EXACT 2026 task (URA Research Group, 2026)	Logic + STEM physics	—	Official task, rules, and evaluation; not a method
CoT / self-consistency (Wei et al., 2022; X. Wang et al., 2023)	Prompt-time reasoning	No	Evaluated after 4B adaptation; no net gain observed
PAL (Gao et al., 2022)	Program-aided arithmetic	No	Python assistance tested; only marginal gain
Logic-LM / LINC (Pan et al., 2023; Olausson et al., 2023)	Logic with solvers	No	Z3 used, but the NL-logic parser limits coverage
DeepSeekMath / Qwen-Math (Shao et al., 2024; Yang et al., 2024)	Maths-specialised LLMs	Yes	Motivation and baselines for low-cost GRPO
LoRA / QLoRA (Hu et al., 2022; Dettmers et al., 2023)	Parameter-efficient tuning	Yes	Applied to a 4B model under a fixed budget
TRA-SAE (this work)	Logic + physics	Yes	Retrieval + SFT + GRPO with reproducible ablations

TABLE 2 | Dataset statistics for EXACT 2026.

Split	Total	Physics	Logic
Train	1,945	1,213	732
Validation	217	141	76
Total	2,162	1,354	808

4.5 | Retrieval-augmented inference and optional components

At inference, the fine-tuned configurations prepend three in-context exemplars retrieved from the training set by dense similarity to the query; the zero-shot configuration uses no exemplars. As additional configurations, two LoRA adapters (one physics-specialised, one logic-specialised) are routed by a TF-IDF and logistic-regression classifier (5,000 n-gram features); when the router confidence exceeds 0.80 the corresponding adapter is selected, otherwise a keyword heuristic is used. Self-consistency voting with N=5 samples is applied as a post-processing step in the full-agent configuration.

5 | Experimental Setup

5.1 | Dataset

Table 2 summarises the data split. The raw sources are a logic corpus and a physics problem set, combined and divided by a 90/10 stratified train/validation split.

Train–validation overlap. Because both splits derive from a single released corpus, we audited the validation set for leakage with three complementary measures: the TF-IDF cosine similarity of each validation question against the full training set, a character 8-gram Jaccard overlap that (unlike word-level features) preserves the numerical values distinguishing otherwise identically worded problems, and inspection of the top-3 retrieval exemplars that cfg0-R actually uses. Surface similarity is high—66 of 217 validation questions reach a maximum TF-IDF cosine ≥ 0.90—but this overwhelmingly reflects template reuse: the corpus instantiates the same physics or logic schema with different numerical values, so near-identical wording maps to a different answer. Under character-level near-identity (8-gram Jaccard ≥ 0.95), only 6 of 217 questions are genuine near-duplicates, and 5 (2.3%) are both near-identical and share the gold answer; three questions appear verbatim in training, one of them carrying a different gold label (a numerical-versus-categorical mismatch of the kind documented in the error analysis). This residual ≈2% overlap is far too small to account for the 17.51 pp headline gain, and the retrieved exemplars are verbatim copies of the validation question in only these three cases (mean top-exemplar cosine 0.68). The audit script (`step10_leakage_check.py`) is released with the code.

5.2 | Baselines and evaluation

Five open-weight 7B-class models (Qwen2.5-Math-7B-Instruct (Yang et al., 2024), Qwen2-Math-7B-Instruct, Llemma-7B (Azerbayev et al., 2024), Mistral-7B-Instruct-v0.3, and DeepSeek-Math-7B-Instruct) and one proprietary model (Gemini-2.5-flash-lite, accessed through an API) are evaluated on the 217 validation samples. The challenge rules prohibit closed-source models in submitted systems (URA Research Group, 2026); the Gemini baseline is therefore reported only as a non-submission diagnostic reference and is not part of the competition-compliant TRA-SAE system, all components of which use open-weight models. To separate format compliance from reasoning capability, each open baseline is evaluated under a fair multi-prompt protocol that tries three answer formats (XML tags, chain-of-thought with a boxed answer, and chain-of-thought with a plain answer line) and reports the best per model. The primary metric is overall accuracy, with physics and logic sub-accuracies reported separately. Statistical significance between configuration pairs is assessed by McNemar's test (two-sided, continuity-corrected), and Wilson score intervals are reported for overall accuracy. Decoding uses greedy generation with a maximum of 1,024 new tokens for all systems. Table 3 lists the principal hyper-parameters.

TABLE 3 | Principal hyper-parameters.

Parameter	Value
Base model	Qwen3.5-4B (BF16)
LoRA r / α / dropout	32 / 64 / 0.05
SFT learning rate / epochs	2×10^{-4} / 3
Logic SFT learning rate / epochs	5×10^{-5} / 2
GRPO learning rate / steps	10^{-6} / 250
GRPO KL coefficient β	0.04
GRPO completions per prompt K	4
Reward weights (w_f, w_c, w_u)	(0.30, 0.60, 0.10)
Length penalty λ	-0.10 if >800 tokens
Retrieval exemplars (fine-tuned)	3
Max new tokens	1,024
GPU	A100-SXM4-80GB

6 | Results and Analysis

6.1 | Main results

Table 4 reports the comparison, and Figure 2 visualises the per-stage accuracy. The strongest configuration, cfg3 (SFT, logic SFT, and GRPO), reaches 47.47% overall accuracy (its increment over the curriculum stage being within validation-set noise), 17.51 pp above the Qwen3.5-4B zero-shot configuration and above all six baselines (Figure 3). The fine-tuned configurations use retrieval-augmented exemplars, whereas the zero-shot configuration and the baselines do not. To separate the two effects, configuration cfg0-R applies the same three-exemplar retrieval to the base model without any fine-tuning: retrieval alone accounts for +9.22 pp (cfg0→cfg0-R, $p < 10^{-3}$), and fine-tuning adds a further +5.07 pp (cfg0-R→cfg1, $p = 0.054$; Table 6). The best configuration also exceeds the reported EXACT 2026 public baseline of 38.71% by 8.76 pp; because that baseline is not reproduced under the present single-pass protocol, it is used only as a contextual reference, and the controlled comparisons rely on cfg0 and the fair 7B-class baselines. Although cfg3 attains the highest overall accuracy, its improvement over cfg2 is driven by physics: logic accuracy (18.42%) does not improve over cfg2 (21.05%), which indicates that the GRPO reward is more effective for answer formatting and numerical reasoning than for formal logical inference. This 2.63 pp difference corresponds to two of 76 logic items and lies within the non-significant cfg2→cfg3 transition ($p = 0.61$, Table 6); it should therefore be read as GRPO leaving logic accuracy unchanged rather than as a genuine regression. The low absolute scores of the maths-specialised 7B baselines should be read in light of the mixed physics–logic format and the strict answer-extraction protocol (Appendix A) rather than as a general ranking of those models: even under the fair best-of-three protocol, the heterogeneous answer space of EXACT 2026 penalises systems that are not adapted to it.

6.2 | Does retrieval explain the baseline gap?

A natural concern is whether the margin over the 7B baselines merely reflects the retrieval exemplars available to the fine-tuned configurations. To test this, the same three-exemplar retrieval used by cfg0-R is supplied to each of the five open-weight baselines under the identical best-of-three protocol (Table 5). Retrieval helps the baselines only marginally on average (+2.6 pp): the strongest baseline rises from 29.03% to 30.88%, and only

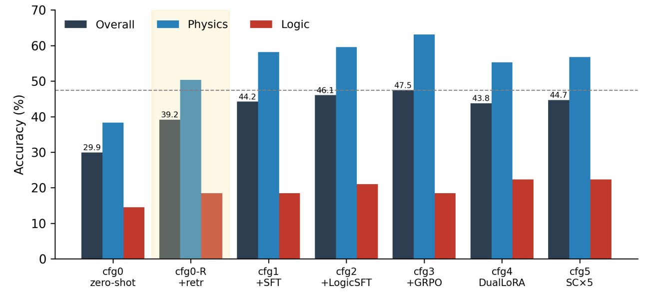

FIGURE 2 | Overall, physics, and logic accuracy across the pipeline stages under a single greedy decoding pass; cfg0-R (shaded) is the retrieval-only control (base model with retrieval, no fine-tuning). The dashed line marks the best overall accuracy (cfg3, 47.47%). Physics accuracy increases monotonically through cfg3, whereas logic accuracy remains below 23% for all configurations.

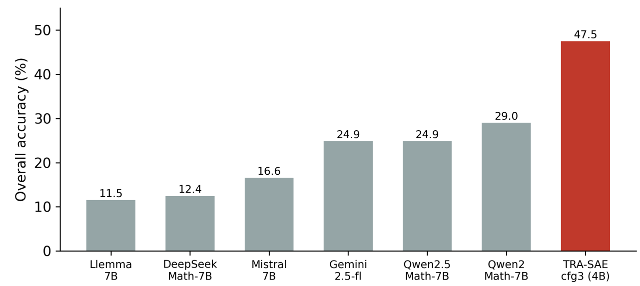

FIGURE 3 | Overall accuracy of the fine-tuned 4B model (cfg3) against the six zero-shot baselines under the fair best-of-three protocol. The fine-tuned model exceeds the strongest baseline by 18.4 pp.

DeepSeek-Math gains substantially (+9.22 pp, from a low base), while one baseline is unchanged and another declines slightly. Crucially, even with identical retrieval the best baseline (30.88%) remains 8.29 pp below the retrieval-only control cfg0-R (39.17%) and 16.59 pp below cfg3 (47.47%). The advantage of TRA-SAE is therefore not attributable to retrieval access: the curriculum-fine-tuned 4B model exploits the same exemplars far more effectively than larger maths-specialised 7B models, consistent with the format-alignment and answer-extraction analysis above.

6.3 | Statistical confidence and significance

Table 6 reports Wilson score intervals and pairwise McNemar tests computed on the single-pass per-sample predictions. The zero-shot-to-SFT (cfg0→cfg1) and zero-shot-to-GRPO (cfg0→cfg3) transitions are both significant; the logic-SFT increment (cfg1→cfg2, $p = 0.39$), the GRPO-over-SFT increment (cfg2→cfg3, $p = 0.61$), and the self-consistency change (cfg3→cfg5, $p = 0.24$) are not significant at the 0.05 level. The overlap among the cfg1–cfg5 intervals indicates that differences among the fine-tuned configurations are small relative to the validation-set size. Because cfg1 includes both supervised fine-tuning and retrieval-augmented exemplars, the cfg0-to-cfg1 comparison should not be read as the isolated effect of fine-tuning; the cfg0-R control separates retrieval from fine-tuning and shows

TABLE 4 | Main results on the EXACT 2026 validation set (217 samples: 141 physics, 76 logic) under a single greedy decoding pass. Baselines report the best of three answer formats. The double dagger (‡) marks McNemar $p < 0.01$ versus cfg0 for the two transitions tested in Table 6 (cfg0→cfg1, cfg0→cfg3).

	System	Overall	Physics	Logic
Zero-shot baselines (best-of-three prompt)				
B1	Qwen2-Math-7B-Instruct	29.03	35.46	17.11
B2	Qwen2.5-Math-7B-Instruct	24.88	34.75	6.58
B3	Gemini-2.5-flash-lite	24.88	32.62	10.53
B4	Mistral-7B-Instruct-v0.3	16.59	13.48	22.37
B5	DeepSeek-Math-7B-Instruct	12.44	14.89	7.89
B6	Llemma-7B	11.52	4.96	23.68
Ours (fine-tuned Qwen3.5-4B)				
cfg0	Zero-shot	29.95	38.30	14.47
cfg0-R	+ Retrieval (no fine-tuning)‡	39.17	50.35	18.42
cfg1	+ SFT‡	44.24	58.16	18.42
cfg2	+ Logic SFT	46.08	59.57	21.05
cfg3	+ GRPO‡	**47.47**	**63.12**	18.42
cfg4	Dual-LoRA + Router	43.78	55.32	22.37
cfg5	Full agent (SC×5)	44.70	56.74	22.37

TABLE 5 | Effect of supplying the cfg0-R retrieval exemplars to the five open-weight baselines (best of three prompt formats, $n = 217$). Even with identical retrieval, the strongest baseline trails the retrieval-only control cfg0-R and the full pipeline cfg3.

Baseline	No retr.	+Retr.	Δ
Qwen2-Math-7B	29.03	30.88	+1.85
Qwen2.5-Math-7B	24.88	27.65	+2.77
Mistral-7B	16.59	16.59	+0.00
DeepSeek-Math-7B	12.44	21.66	+9.22
Llemma-7B	11.52	10.60	−0.92
cfg0-R (ours, retrieval only)	—	*39.17*	
cfg3 (ours, full pipeline)	—	*47.47*	

that retrieval accounts for the larger share of the first-stage improvement.

6.4 | Multi-seed reliability

Table 7 reports cfg3 re-evaluated with three generation seeds. Under the single-pass protocol the overall standard deviation is 0.70 pp, which is small relative to the 17.51 pp zero-shot-to-cfg3 gain and confirms that the reported accuracy is stable across seeds.

6.5 | External generalisation: physics and logic

To assess transfer beyond the in-domain split, cfg0, cfg2, and cfg3 are evaluated on the MMLU (Hendrycks et al., 2021) high-school and college physics subsets (253 four-option multiple-choice questions) without additional fine-tuning (Table 8). The fine-tuned configurations improve over the zero-shot model by up to 6.33 pp, with monotonic improvement across stages, which provides an external sanity check that the in-domain curriculum transfers rather than overfitting to the EXACT 2026 format. These results are not a state-of-the-art claim on MMLU.

The same transfer test is repeated for logic on FOLIO (Han et al., 2024), a first-order-logic entailment benchmark whose three-way label space (entailed / contradicted / undetermined) matches the EXACT 2026 logic format (Table 9). The zero-shot base model scores at the majority-class level (34.48%), whereas the logic-curriculum configuration cfg2 reaches 48.28%—a +13.80 pp gain—and cfg3 attains 46.80%. The logic curriculum therefore induces genuine, transferable logical reasoning rather than overfitting to the EXACT 2026 answer format, and the small cfg2→cfg3 decrease (−1.48 pp) mirrors the in-domain result that GRPO does not improve logic. The larger logic transfer (+13.80 pp) than physics transfer (+6.33 pp) indicates that the curriculum's logic stage, not only its physics exposure, carries beyond the competition split.

6.6 | Tool-augmented reasoning

Table 10 compares direct inference with a tool-augmented variant of the best checkpoint, in which a Python evaluator handles physics arithmetic and the Z3 solver handles logic inference; the model answer is retained when the solver abstains. Both arms use a single-pass protocol *without* retrieval exemplars, so the absolute values are not directly comparable to Table 4; the purpose is to isolate the tool effect. Tool augmentation yields a marginal overall change (+0.46 pp), with a small physics gain from the Python evaluator and a small logic decrease. The Z3 solver abstained on the natural-language logic items because their premises were not reliably parseable into a formal specification, which limits the benefit of symbolic tooling under the current parser.

6.7 | Reward component ablation

Table 11 reports four reward variants trained for 150 GRPO steps from the Phase-1 checkpoint and evaluated under single-pass decoding without retrieval. The correctness-only variant (R4) attains the highest accuracy at this budget, 1.84 pp above the full mixture (R1); removing the format term (R3) reduces physics accuracy by 2.84 pp. At this 150-step budget the auxiliary unit-consistency and format terms do not provide a net accuracy benefit relative to a correctness-only objective. A plausible explanation is that the correctness reward supplies the most direct training signal and that auxiliary terms add gradient variance that

TABLE 6 | Wilson 95% intervals (left) and pairwise McNemar tests (right) under the single-pass protocol ($n = 217$). For a pair $A \to B$, b is the count where B is correct and A is wrong, and c is the reverse.

System	Overall (95% CI)	Pair	ΔAcc	b	c	p
cfg0	29.95 [24.3, 36.4]	cfg0→cfg0-R	+9.22	26	6	7.8×10^{-4}
cfg0-R	39.17 [32.9, 45.8]	cfg0-R→cfg1	+5.07	19	8	0.054
cfg1	44.24 [37.8, 50.9]	cfg0→cfg3	+17.51	43	5	9.3×10^{-8}
cfg2	46.08 [39.6, 52.7]	cfg1→cfg2	+1.84	8	4	0.39
cfg3	47.47 [40.9, 54.1]	cfg2→cfg3	+1.38	9	6	0.61
cfg4	43.78 [37.3, 50.4]	cfg3→cfg5	−2.76	6	12	0.24
cfg5	44.70 [38.2, 51.4]					

TABLE 7 | cfg3 multi-seed overall accuracy (single greedy pass).

Seed	Overall accuracy (%)
42	47.47
1337	46.08
2024	46.54
Mean ± std	46.70 ± 0.70

TABLE 8 | Zero-shot transfer to MMLU physics (253 questions: 151 high-school, 102 college).

Configuration	Overall	HS	College
cfg0 (zero-shot)	42.29	41.06	44.12
cfg2 (+ logic SFT)	47.04	47.68	46.08
cfg3 (+ GRPO)	**48.62**	46.36	51.96

TABLE 9 | Zero-shot transfer to FOLIO logic (203 validation items; majority-class baseline ≈ 35.5%).

Configuration	Accuracy	Δ vs cfg0
cfg0 (zero-shot)	34.48	—
cfg2 (+ logic SFT)	**48.28**	+13.80
cfg3 (+ GRPO)	46.80	+12.32

is not amortised within 150 steps; whether the ordering reverses at longer horizons was not evaluated within the compute budget.

TABLE 10 | Direct versus tool-augmented inference on the best checkpoint ($n = 217$, single pass, no retrieval exemplars). Tool activations: Python evaluator on 62 samples; Z3 abstained on the remaining cases.

Arm	Overall	Physics	Logic
A: direct	39.63	49.65	21.05
B: + Python / Z3 tools	40.09	51.06	19.74
Δ (B − A)	+0.46	+1.41	−1.31

TABLE 11 | Reward component ablation (150 GRPO steps from the Phase-1 checkpoint, single pass, no retrieval, $n = 217$).

	Reward configuration	Overall	Physics
R1	Full (format, correct, unit, length)	39.63	49.65
R2	No unit term	39.17	49.65
R3	No format term	38.25	46.81
R4	Correctness only	**41.47**	**51.06**

TABLE 12 | Compute profile on one A100-SXM4-80GB at US$3.67 per GPU-hour.

Stage	Time (min)	Cost (US$)
SFT Phase 1	58.8	3.60
Logic SFT (Phase 1.5)	15.7	0.96
GRPO Phase 2 (mixed)	79.8	4.89
GRPO Phase 2 (physics)	76.6	4.69
GRPO Phase 2 (logic)	62.9	3.85
Total training	293.8	**18.0**

6.8 | Compute cost

Table 12 summarises the compute profile. Phase-1 SFT provides the highest accuracy per training dollar, and the full pipeline cost is below one A100-day.

6.9 | Error analysis

Because the canonical evaluation retained only resolved outputs, a dedicated single-pass run of cfg3 was performed to retain the full generations required for error classification. Table 13 reports a rule-based error taxonomy on that run ($n = 117$ incorrect predictions; the run scores 46.08% overall, within the seed variation of Table 7). The largest category is answer-extraction failure (72.6%): the response does not yield an answer that the automated verifier can reconcile with the ground truth, and inspection indicates that many of these are incomplete generations that terminate before a final answer is emitted. The second category is logical-fallacy and quantifier error (17.9%), concentrated in the logic subset. Two observations follow. First, because a large fraction of measured errors are extraction or generation-length failures rather than substantive reasoning errors, part of the measured error may be attributable to extraction limitations rather than to reasoning; a manual annotation protocol is required to quantify this fraction, which is left to future work. Second, among substantive reasoning errors the logical-fallacy category dominates, consistent with the persistent physics–logic accuracy gap and identifying formal logic as the principal bottleneck at this scale.

To make these categories concrete, Table 14 lists representative cfg3 failures drawn from the single-pass run. The dominant pattern is an answer-extraction mismatch in which the model reasons soundly but emits an answer whose form the verifier cannot reconcile with a categorical ground truth, together with truncated

TABLE 13 | Rule-based error taxonomy for incorrect cfg3 predictions ($n = 117$, single-pass run). Labels are produced by an automated classifier; manual validation is left to future work.

ID	Error type	Count	%
E6	Answer-extraction / incomplete output	85	72.6
E4	Logical fallacy / quantifier error	21	17.9
E0	Unclassified	9	7.7
E1	Unit or dimension error	2	1.7

generations that stop before a final answer is emitted; the substantive logical failures are typically faulty quantifier or existential reasoning.

7 | Discussion

7.1 | Self-consistency at 4B scale

Self-consistency is effective when errors are independent across samples (X. Wang et al., 2023). The present error profile indicates that a large share of failures are systematic, either fixed answer-format failures or repeated quantifier errors, which majority voting reinforces rather than corrects. This accounts for the non-significant decrease from cfg3 to cfg5 (-2.76 pp, $p = 0.24$) despite five times the inference compute.

7.2 | Domain routing

The TF-IDF and logistic-regression router classifies clean physics and logic items reliably, but border-zone questions, such as physics problems phrased as logical conditionals, are routed to the wrong adapter. The Dual-LoRA configuration (cfg4, 43.78%) does not exceed the single mixed adapter (cfg3, 47.47%), which indicates that routing noise offsets the benefit of domain specialisation at this scale.

7.3 | Limitations

- *Validation-set size.* With 217 samples, Wilson intervals span roughly ± 7 pp; differences below approximately 5 pp are not statistically separable, including the GRPO-over-SFT increment ($+1.38$ pp, $p = 0.61$). The 217 samples are a 90/10 stratified split of the released training data; the official competition test labels were not available, so no independent test-set result is reported. Consequently only the large cfg0→cfg3 gain ($p = 9.3 \times 10^{-8}$) survives the small-sample caveat, whereas the increments among the fine-tuned configurations do not.
- *Inference asymmetry.* The fine-tuned configurations use three retrieved in-context exemplars, whereas the baselines do not. Configuration cfg0-R (Table 4) isolates the retrieval contribution from fine-tuning; cross-family comparisons with the zero-shot baselines should nonetheless be read with the retrieval component in mind.
- *Extraction-bounded measurement.* A large fraction of residual errors are answer-extraction or generation-length failures; a manual annotation study is required to separate these from substantive reasoning errors. The rule-based taxonomy is automatically assigned and should be read as diagnostic rather than definitive. In particular, no human or expert annotation of the error categories was performed; quantifying the precise share of genuine reasoning failures would require such a study, which we leave to future work.
- *Budget-conditional reward finding.* The advantage of the correctness-only reward holds at a 150-step budget and may not persist at longer horizons.
- *Single in-domain benchmark.* The primary results use one competition split; the MMLU evaluation is an external sanity check rather than a broad generalisation study.
- *Logic gap.* The physics–logic gap and the logical-fallacy error rate indicate that standard SFT and GRPO are insufficient for formal symbolic logic at 4B scale; step-level supervision (Lightman et al., 2023; P. Wang et al., 2024) or solver-guided training are the most direct avenues for improvement. Because the measured gains are driven primarily by the physics domain while logic accuracy remains below 23%, TRA-SAE should be read as a low-cost empirical study of what curriculum and retrieval achieve at this scale rather than as a complete solution to mixed-domain reasoning.

8 | Conclusion

This paper presented TRA-SAE, a low-cost curriculum training pipeline for mixed-domain scientific reasoning that combines curriculum SFT, GRPO with lightweight reward shaping, retrieval-augmented inference, and an optional Dual-LoRA routing agent. Under a uniform single-pass protocol, the strongest configuration reaches 47.47% overall accuracy on the EXACT 2026 validation set, a 17.51 pp improvement over the Qwen3.5-4B zero-shot configuration, using a 4B open model and under US$64 of total compute. Across the three training stages, the initial supervised fine-tuning delivers the highest accuracy gain per training dollar (Table 12); together with training-free retrieval, it recovers most of the attainable gain for under US$4 of fine-tuning, whereas the later GRPO stages add compute without a statistically separable benefit. A controlled decomposition attributes $+9.22$ pp of the improvement to retrieval-augmented exemplars and a further $+5.07$ pp to supervised fine-tuning, while the GRPO-over-SFT increment is not statistically separable at the validation-set size. A fair multi-prompt re-evaluation places the 7B-class baselines at 11.5–29.0% and supplying them with the same retrieval does not close the gap, external evaluations show positive transfer on MMLU physics ($+6.33$ pp) and FOLIO logic ($+13.80$ pp over zero-shot), and a tool-augmented variant produces only a marginal change. The error analysis indicates that answer-extraction and generation-length failures account for a large share of residual errors and that formal logic is the principal reasoning bottleneck. Future work includes a manual error-annotation study, a unified single-protocol re-evaluation that removes the retrieval asymmetry, scaling to 7B and 14B checkpoints, cross-dataset evaluation on SciBench (X. Wang et al., 2024) and GPQA (Rein et al., 2023), and tighter integration of solver-guided training signals for the logic domain.

TABLE 14 | Representative cfg3 errors (single-pass run) illustrating the dominant categories of Table 13.

Category	Symptom (abridged)	Interpretation
E6: extraction mismatch	Capacitor charge after disconnection: the model answers “100 μC” (the correct value), but the ground truth is the categorical label “unchanged”	Sound reasoning scored wrong; numerical versus categorical answer space
E6: incomplete output	Electric-field problem: the generation terminates mid-derivation (“… A is at $x = 1$ cm. B is”) with no final answer	Generation-length limit; constrained decoding could recover the answer
E4: logical fallacy	Quantifier item: the model declares the query a tautology and answers “Yes” where the entailment is “No”	Faulty existential/quantifier reasoning, not an extraction issue

Author Contributions

The author designed the study, implemented the pipeline, conducted the experiments, analysed the results, and wrote the manuscript.

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Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The source code, evaluation scripts, trained LoRA adapters, and result artefacts that support the findings of this study are available at https://github.com/ANONYMISED-FOR-REVIEW (a permanent archive will be deposited on acceptance). The EXACT 2026 competition data are governed by the challenge access policy and are not redistributed here; the MMLU benchmark used for external evaluation is publicly available from its original source.

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APPENDIX

A | Prompt Templates and Answer Extraction

The baseline systems in Table 4 are evaluated under a fair multi-prompt protocol with three answer formats, and the best score per model is reported. Each prompt is supplied through the native chat template of the corresponding model, as a system message followed by a user message containing the question. The three system prompts are reproduced below.

```
You are an expert in Logic and Physics. Think step by
     step and
respond in the EXACT format:
<reasoning>
[Your step-by-step reasoning]
</reasoning>
<answer>
[Final answer: letter / Yes/No/Unknown / number+unit]
</answer>
<explanation>
[Concise explanation]
</explanation>
```

Listing 1 XML prompt.

```
You are an expert in Logic and Physics. Solve the problem
     step by
step. At the end, write your final answer inside \boxed
     {...}.
For MCQ, write the letter. For yes/no questions, write
     Yes, No, or
Unknown. For numerical problems, include units.
```

Listing 2 Boxed chain-of-thought prompt.

```
You are an expert in Logic and Physics. Solve the problem
     step by
step, showing all reasoning. End your response with:
'The answer is: [your answer]'
For MCQ, the answer is just the letter (A/B/C/D). For yes
     /no
questions, answer Yes, No, or Unknown. For numerical
     problems,
include units.
```

Listing 3 Plain chain-of-thought prompt.

A single extractor is applied to every response in the following priority order: (i) the content of an `<answer>...</answer>` tag; (ii) the content of a `\boxed{...}` expression; (iii) the text following an "Answer:" or "The answer is:" marker; and (iv) the last non-empty line. Physics answers are compared after SI-unit normalisation with a 5% relative tolerance, and logic answers are compared by normalised string matching, with the Z3 solver invoked when a parseable formal representation is available.